

Template for DATA 501 W26 – Final Project Report

Title Page

- Include project title, student names and IDs, date, instructors, and course name.
NOTE: make sure the title of your project is detailed and includes your project type (e.g., prediction, clustering, anomaly detection)

Table of Contents

Abstract (5 points)

- 200-300 words
NOTE: High-level overview of your project to be understandable to the general public.

Chapter 1: Introduction (15 points)

- Background information and motivation
- Overall aim (if applicable) + research questions
NOTE: must define the scope clearly and in detail with correct in-text citation

Chapter 2: Methods (30 points)

2.1. Data Description

- Data source(s) with an explanation of how the data was collected
- Size, structure, and format of data
 - Include data integrity issues if they exist (e.g., missing data)
- Key variables and features
- Ethical considerations (if applicable)

2.2. Cleaning and Pre-Processing

- The steps taken to clean the data
- Feature engineering or transformations used (include reasoning for all data transformation techniques)
- Tools/libraries used

2.3. Exploratory Data Analysis

- Visualizations/stat summaries to explore patterns
- NOTE 1: Ensure data visualizations are consistent and provide visual context to the story. What is the purpose of a certain data visualization being included in the report?
- NOTE 2: Any insights and changes to the hypotheses you discovered from EDA should be mentioned in the RESULTS, not here.

2.4. Modeling/Analytical Approach

- Description of the models and metrics used
- Include the rationale for choosing your methods

Chapter 3: Results (30 points)

- Results should be in numerical form with verbal and unbiased explanations of what they represent (NO INTERPRETATIONS). Include appropriate figure and table captions when presenting your findings.

NOTE: Ensure that such results are justified by code/formulae, which should appear in an appropriate Appendix section.

- If applicable, compare multiple methods and include evaluation metrics in table/figure format

Chapter 4: Discussion (30 points)

- You should include future directions at the end
- Interpret your results (should reflect on literature as well)
- Limitations of the analysis
- If applicable, explicitly address how their findings could inform decision-making or policies.

Chapter 5: Conclusion (5 points)

References (10 points)

- Use APA format

Appendix (if applicable)

NOTE: all computations/code should be sub-sectioned in the Appendix)

Writing Quality and Formatting will also be marked (15 points)

All written portions should be in size 12 font (apart from titles and figure/table captions), 2.54 margins, and one-column pages.

Presentation Rubric (Total: 30 Points)

Criteria	Excellent (5 pts)	Good (3–4 pts)	Needs Improvement (1–2 pts)
Content & Clarity	Clear, concise; all content is well-organized	Mostly clear; minor issues in flow	Unclear or lacks structure
Understanding of Project	Demonstrates deep understanding	Understands main ideas	Limited understanding or incorrect explanations
EDA & Model Justification	Thoughtful analysis and rationale	Mostly appropriate choices	Weak analysis or unclear justification
Visual Aids	Clear, relevant, easy to read	Mostly clear; minor formatting issues	Cluttered or difficult to interpret visuals
Delivery	Engaging, confident, excellent pacing	Some nervousness or pacing issues	Monotone, rushed, or hard to hear
Q&A Handling	Responds thoughtfully and correctly	Handles most questions well	Struggles to answer or avoids questions